

Gianmarco La Rosa

Curriculum Vitae

Palermo, Italy

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EDUCATION

- 2024 **Ph.D. in Mathematics and Computational Sciences (and Doctor Europaeus)**, *University of Palermo*, Palermo, Italy
Thesis: On nilpotent Leibniz algebras, Lie biderivations, and related topics.
- 2020 **Master's Degree in Mathematics (LM-40)**, *University of Palermo*, Palermo, Italy
Final grade: 110/110 cum laude. Thesis: Extensions of Lie algebras.
- 2018 **Bachelor's Degree in Mathematics (L-35)**, *University of Palermo*, Palermo, Italy
Final grade: 102/110. Thesis: Modules over principal ideal domains.

ACADEMIC POSITIONS

POSTDOC

- 06/2024–
present **Research Fellow (Borsista di Ricerca)**, *University of Palermo*, Palermo, Italy
Title: Mathematical Models and Knowledge Representation Under Uncertainty for Explainable Artificial Intelligence with a Geothematic Focus.
- 05/2024–
05/2026 **Research Fellow (Borsista di Ricerca)**, *University of Palermo*, Palermo, Italy
Title: Mathematical models for explainable AI using fuzzy ontologies and categorical approaches.

TEACHING

- 02/2024–
present **Adjunct Professor (Docente a contratto)**, *University of Palermo*, Palermo, Italy
Teaching Geometry for the Marine Technologies Engineering.
- 02/2024–
04/2026 **Adjunct Professor (Docente a contratto)**, *University of Palermo*, Palermo, Italy
Teaching Geometry for the Sustainable Development Engineering program.
- 2021–2022 **University Tutor**, *University of Palermo*, Palermo, Italy
Worked on courses in Mathematical Analysis and Geometry for various engineering programs.
- 2020–2021 **University Tutor**, *University of Palermo*, Palermo, Italy
Courses in Geometry and Fundamentals of Mathematics for engineering and optics students.
- 2019–2020 **University Tutor**, *University of Palermo*, Palermo, Italy
Conducted exercises for courses in Geometry and Algebra (B.Sc. in Physics).

- 02/2020– **School Tutor**, *The Math House*, Palermo, Italy
- 11/2020 Provided tutoring in Mathematics and Physics for secondary school and university students.
- VISITING
- Apr–May Visiting researcher at Instituto de Matemática Pura e Aplicada (IMPA), Rio de
2026 Janeiro (Brazil), under the supervision of Prof. Henrique Bursztyn. Research focus: Courant-Dorfman algebras and Courant algebroids
- Feb 2026 Visiting researcher at Georgian Technical University, Tblisi (Georgia), under the supervision of Prof. Vakhtang Kvaratskhelia
- Feb–May Visiting researcher at City, University of London, London (United Kingdom), un-
2023 der the supervision of Prof. Markus Linckelmann. Research focus: Homological Algebra.
- Oct 2023 Visiting researcher at the University of Szeged, Szeged (Hungary), hosted by Prof. P. G. Nagy. Research focus: C^* -algebras and midpoint operators.

RESEARCH INTERESTS

- Algebra Non-associative algebras (Lie algebras, Leibniz algebras), derivations of non-associative algebras and related topics (e. g. biderivations)
- Geometry Lie theory, smooth manifolds, Lie groups, Lie racks, coquecigrue problem
- Logic Fuzzy set theory, fuzzy logic, and fuzzy abstract algebra (e. g. fuzzy vector spaces, fuzzy Lie algebras)
- Knowledge Building ontologies for AI implementations (e. g. Natural language processing),
Representa- fuzzy reasoning in crisp ontologies (e. g. FuzzyOWL 2)
tion

PUBLICATIONS

PREPRINTS

- 2026 *Biderivations of Lie algebras and field extensions* (with A. Di Bartolo and F. P. Di Fatta)
- 2026 *A geometric point of view on biderivations* (with A. Dioguardi Burgio and M. Galici)
- 2026 *Fuzzy Nilpotent Lie Algebras: Bases, Isomorphisms, and a Measure of Nilpotency* (with G. Filippone, M. Galici, F. Piazza, and M. E. Tabacchi), 10.20944/preprints202604.0066.v1
- 2025 *An Application of Fuzzy Set Theory and Interval-Based Distances for Expert Judgment Aggregation* (with G. Failla and M. E. Tabacchi)
- 2025 *On Lie-holomorphs of Leibniz algebras* (with M. Mancini), arXiv:2512.1927
- 2025 *Biderivations of complete Leibniz algebras* (with A. Di Bartolo and F. P. Di Fatta), arXiv:2510.16426
- 2025 *A Lie Bracket on the Space of (Right) Biderivations of a Lie Algebra* (with A. Di Bartolo), arXiv:2507.12089

JOURNAL ARTICLES

- 2026 *A new definition of superbiderivations for Lie superalgebras*, Journal of Algebra and Its Applications, 2750195 (with A. Di Bartolo and F. P. Di Fatta), 10.1142/S0219498827501957
- 2026 *Computing with Speculations*, to appear in Soft Computing, Springer (with G. Filippone and M. E. Tabacchi)
- 2026 *Logic and Probabilistic Operations on a Decision Matrix in a Fuzzy Multi-Criteria Decision-Making Problem*, Mathematics, 14(5), 778 (with L. Castronovo, G. Filippone, G. Sanfilippo, and M. E. Tabacchi), 10.3390/math14050778
- 202? *Flag Manifolds without One Codimensional Strata* (with A. Di Bartolo), to appear in Differential Geometry of Manifolds of Figures
- 2025 *Fuzzy MCGDM Approach for Ontology Fuzzification*, Electronics, 14(18), 3596 (with L. Castronovo, G. Filippone, M. Galici, and M.E. Tabacchi), 10.3390/electronics14183596
- 2025 *On a class of two-step solvable Leibniz algebras*, International Journal of Algebra and Computation, 2025, vol. 35, No. 05, p. 733-758 (with A. Di Bartolo), 10.1142/S0218196725500249
- 2024 *Non-nilpotent Leibniz algebras with one-dimensional derived subalgebra*, Mediterranean Journal of Mathematics, 2024 (with A. Di Bartolo and M. Mancini), 10.1007/s00009-024-02679-0
- 2024 *Isotopisms of nilpotent Leibniz algebras and Lie racks* (with M. Mancini and G.P. Nagy), Communications in Algebra, 2024, doi.org/10.1080/00927872.2024.2330686
- 2023 *Biderivations of complete Lie algebras* (with A. Di Bartolo), Journal of Algebra and Its Applications, 2023, 10.1142/S0219498825500161
- 2023 *Derivations of two-step nilpotent algebras* (with M. Mancini), Communications in Algebra, 2023, 10.1080/00927872.2023.2222415
- 2022 *Two-step nilpotent Leibniz algebras* (with M. Mancini), Linear Algebra and its Applications, 2022, 10.1016/j.laa.2021.12.013

CONFERENCE PAPERS

- 2026 *Functors Between Fuzzy Varieties* (with F. Piazza), in *Information Processing and Management of Uncertainty in Knowledge-Based Systems*, ed. by B. Vantaggi, G. Coletti, T. Denoeux, A. Laurent, D. Petturiti, E. Miranda, J. Medina, B. Bouchon-Meunier and R. R. Yager, Cham: Springer Nature Switzerland, 2026, pp. 291–304, ISBN 978-3-032-28997-1
- 2026 *Fuzzy Rule-Based Approach for Weighting Artificial Experts Involved in a Multi-Criteria Group Decision-Making Problem* (with L. Castronovo, G. Filippone and M. E. Tabacchi), in *Information Processing and Management of Uncertainty in Knowledge-Based Systems*, ed. by B. Vantaggi, G. Coletti, T. Denoeux, A. Laurent, D. Petturiti, E. Miranda, J. Medina, B. Bouchon-Meunier and R. R. Yager, Cham: Springer Nature Switzerland, 2026, pp. 277–290, ISBN 978-3-032-28997-1

- 2026 *SDF FuzzIA: A Fuzzy-Based, AI Support System for Decision-Making Frameworks* (with L. Castronovo, G. Filippone, M. Galici, A. M. Pavone, and M. E. Tabacchi), Decision Sciences. DSA ISC 2025, Lecture Notes in Computer Science, vol. 16405, Springer, Cham, 2026, doi: 10.1007/978-3-032-21811-7_28
- 2025 *A new definition of fuzzy nilpotent Lie algebras*, 2025 IEEE International Conference on Fuzzy Systems (FUZZ), Reims, France, 2025, pp. 1-5 (with G. Filippone, M. Galici, F. Piazza, and M.E. Tabacchi), 10.1109/FUZZ62266.2025.11152208
- 2025 *Ontology Aggregation with Maximum Consensus Based on a Fuzzy Multi-criteria Group Decision-Making Method*, Advances in Fuzzy Logic and Technology. EUSFLAT 2025. Lecture Notes in Computer Science, vol 15884. Springer, Cham. (with L. Castronovo, G. Filippone, M. Galici, and M.E. Tabacchi), 10.1007/978-3-031-97228-7_7
- 2025 *Some Results on Fuzzy Basis of Fuzzy Lie Algebras*, Advances in Fuzzy Logic and Technology. EUSFLAT 2025. Lecture Notes in Computer Science, vol 15884. Springer, Cham. (with G. Filippone, M. Galici, and M.E. Tabacchi), 10.1007/978-3-031-97228-7_9
- 2024 *Fuzziness and Lie Algebras*, Information Processing and Management of Uncertainty in Knowledge-Based Systems: 20th International Conference, IPMU 2024, Lisbon, Portugal, July 22-26, 2024, Proceedings. Vol. 1175. Lecture Notes in Networks and Systems. Springer, 2024 (with G. Filippone, M. Mancini, and M.E. Tabacchi), 10.1007/978-3-031-74000-8_10
- 2024 *Max Black and a Logic of Vagueness*, Information Processing and Management of Uncertainty in Knowledge-Based Systems: 20th International Conference, IPMU 2024, Lisbon, Portugal, July 22-26, 2024, Proceedings. Vol. 1175. Lecture Notes in Networks and Systems. Springer, 2024 (with G. Filippone and M.E. Tabacchi), https://link.springer.com/chapter/10.1007/978-3-031-74000-8_9
- 2024 *SDF-FuzzIA: A Fuzzy-Ontology based plug-in for the intelligent analysis of geo-thematic data*, Scalable Uncertainty Management. SUM 2024. Lecture Notes in Computer Science, vol 15350. Springer, 2024 (with G. Filippone and M.E. Tabacchi), 10.1007/978-3-031-76235-2_13

BOOK CHAPTERS

- 2026 *Fuzzy Ontology-based Explainable Artificial Intelligence for the Geo-Thematic Field*, accepted for a FM Square Foundation's book, Springer, 2026 (with G. Filippone, M. Galici, G. Giacomelli, M. Mancini, M. E. Tabacchi, and D. Tegolo)

PHD THESIS

- 2024 On nilpotent Leibniz algebras, Lie biderivations and related topics, <https://iris.unipa.it/handle/10447/623814>

CONFERENCES AND SEMINARS

ORGANIZATION

- 2026 21st International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems (IPMU2026), Rome, Italy, organiser of the special session "Multi-valued Logics: theoretical and practical aspects"
- 2026 5th Italian Meeting on Probability and Mathematical Statistics Uncertainty (IMPMS26), Palermo, Italy, organiser of the special session "Vagueness, and Decision Support: Logical and AI Approaches"
- 2024 2nd AMS-UMI International Joint Meeting, Palermo, Italy. Role: Participant and Staff
- 2024 SUM 2024 - The 16th International Conference on Scalable Uncertainty Management, Palermo, Italy. Role: Author and Local Committee
- 2023 PSSL108 - 108th Peripatetic Seminar on Sheaves and Logic (Satellite Event: Logic & Algebra), Terrasini, Italy. Role: Local Committee

TALKS

- 2026 21st International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems (IPMU2026), Rome, Italy. *Functors between Fuzzy Varieties*
- 2025 EUSFLAT 2025, 14th Conference of the European Society for Fuzzy Logic and Technology, Riga, Latvia. *Some Results on Fuzzy Basis of Fuzzy Lie Algebras*
- 2024 Groups and Topological Groups, Rostock, Germany. *Biderivations of complete Lie algebras*
- 2022 Groups and Topological Groups, Berlin, Germany. *Classification and derivations of two-step nilpotent Leibniz algebras*
- 2022 Seminar Sophus Lie, Nordfjordeid, Norway. *Two-step nilpotent Leibniz algebras*

PARTECIPATIONS

- 2024 ARTA24 - Advances in Ring Theory and Applications, Online
- 2023 Scientific Thought in Animation and Comics, Palermo, Italy. Scientific presentation: "The mathematical artistry behind animation: a journey from Geometry to Algebra"
- 2023 Groups and Topological Groups, Milan, Italy
- 2023 London Algebra Colloquium, London, UK
- 2022 Workshop on Associative Rings and Algebra with Additional Structures, Online
- 2022 5th Workshop in Categorical Algebra, Gargnano del Garda, BS, Italy
- 2021 CT20→CT21, Category Theory 2020, Online
- 2021 CaCS2021 - Categories and Companions Symposium 2021, Online

GRANTS & PROJECTS

- 2024-present Research scholarship, SDF Project, University of Palermo
- 2026 Research Grant, *EUROSTART – Università degli Studi di Palermo*, €9.300. Project: *Q-Struct: Quantum Reasoning on Algebraic Structures* (Participant). PI: A. M. Pavone.

- 2026 Funding from Instituto de Matemática Pura e Aplicada (IMPA), Rio de Janeiro, Brazil, for a visiting period of research at IMPA.
- 2026 Funding from INdAM (Istituto Nazionale di Alta Matematica) for a visiting period of research at Instituto de Matemática Pura e Aplicada (IMPA), Rio de Janeiro, Brazil.
- 2026 Research Project Grant, *INdAM–GNSAGA*, €1.800. Project: *Flag-Transitive 2-Designs, Loops and Non-associative Algebras*. CUP E53C25002010001.
- 2020–2024 PhD scholarship, Doctoral Program in Mathematics and Computational Sciences, University Consortium of Palermo, Catania, and Messina.
- 2024 Funding from PhD program to attend and present a talk at the Groups and Topological Groups, Rostock, Germany.
- 2024 Funding from INDAM¹ to attend and present a talk at the Groups and Topological Groups, Rostock, Germany.
- Feb–May 2023 PhD scholarship increase for a research stay abroad at City, University of London (UK), under the supervision of Prof. Markus Linckelmann. Research topic: Homological Algebra.
- Oct 2023 PhD scholarship increase for a short research stay at the University of Szeged (Hungary), hosted by Prof. P. G. Nagy. Research topic: C^* -algebras and midpoint operators.
- 2023 Funding from PhD program to attend the Groups and Topological Groups, Milan, Italy.
- 2023 Funding from INDAM to attend the Groups and Topological Groups, Milan, Italy.
- 2022 Funding from PhD program to attend the 5th Workshop in Categorical Algebra, Gargnano del Garda (BS), Italy.
- 2022 Funding from PhD program to attend and present a talk at the Seminar Sophus Lie, Nordfjordeid, Norway.
- 2022 Funding from PhD program to attend and present a talk at the Groups and Topological Groups, Berlin, Germany.
- 2022 Funding from The Free University of Berlin to attend and present a talk at the Groups and Topological Groups, Berlin, Germany.

SKILLS

- Languages Italian (Native), English (B2).
- Technical Python, C/C++, MATLAB, Wolfram Mathematica, OWL Ontology Development, Maple.

REFeree ACTIVITIES

¹Istituto Nazionale di Alta Matematica

Acta Scientiarum Mathematicarum, Advances in Mathematical Physics, Applied Sciences, Communications in Algebra, European Journal of Mathematics, IEEE Computational Intelligence Magazine, International Journal of Mathematics and Mathematical Sciences, Journal of Algebra Combinatorics Discrete Structures and Applications, Journal of Nonlinear Mathematical Physics, Lecture Notes in Artificial Intelligence, Lecture Notes in Computer Science, Mathematics, Sustainability

MISCELLANEOUS

Hobbies Guitar player, Basketball, and Member of the Italian Alpine Club (CAI) since 2023.

Licenses Driving Licenses A and B.

REFERENCES

Prof. Giovanni Falcone, Full Professor at the University of Palermo, Italy, giovanni.falcone@unipa.it

Prof. Gábor Peter Nagy, Full Professor at the University of Szeged, Hungary, nagyg@math.u-szeged.hu

Prof. Marco Elio Tabacchi, Associate Professore at the University of Palermo, Italy, marcoelio.tabacchi@unipa.it

Prof.ssa Elisabetta Tornatore, Associate Professor at the University of Palermo, Italy, elisa.tornatore@unipa.it